Q1. [25 pts] Maximum Likelihood Estimation

There are 5 balls in a bag. Each ball is either red or blue. Let θ (an integer) be the number of blue balls. We want to estimate θ, so we draw 4 balls with replacement out of the bag, replacing each one before drawing the next. We get "blue," "red," "blue," and "blue" (in that order).

(a) [10 pts] If θ is fixed, what is the likelihood of getting exactly that sequence of colors (expressed as a function of θ)?

(b) [10 pts] Draw a table such as one showing below (as a fraction) the likelihood of getting exactly that sequence of colors, for every value of θ from zero to 5 inclusive.

θ	$\mathcal{L}(\theta; \langle$ blue, red, blue, blue $\rangle)$
0	?
1	?
2	?
3	?
4	?
5	?

(c) [5 pts] What is the maximum likelihood estimate for θ? (Chosen among all integers; not among all real numbers.)

Q1.

(a)

$$\mathcal{L}(\theta; X) = P(X; \theta) = \left(\tfrac{5-\theta}{5}\right)\left(\tfrac{\theta}{5}\right)^3.$$

(b)

θ	$\mathcal{L}(\theta; \langle$ blue, red, blue, blue $\rangle)$
0	0
1	4 / 625
2	24 / 625
3	54 / 625
4	64 / 625
5	0

(c)

The maximum likelihood estimate for θ is 4.

Q3. [25 pts] Naive Bayes Classifier

Consider the following data set with three features X_1, X_2, X_3, and label Y.

X_1	X_2	X_3	Y
1	0	1	0
0	0	1	0
1	1	0	0
0	0	0	1
1	1	1	1
1	0	0	1

(a) [10 pts] Suppose that we wish to fit a Naive Bayes model to the data $P(Y|X_1, \ldots, X_n) \propto P(Y)\prod_{i=1}^{n} P(X_i|Y)$. Calculate the probability $P[Y = 0|X = (0,1,1)]$. Show your work.

(b) [15 points] Given the same data set, apply Laplace smoothing with $k = 1$. Calculate $P[Y = 1|X = (0,1,0)]$.

Q3.

(a) $P[Y=y|X_1=x_1, X_2=x_2, X_3=x_3] = P[Y=y] \cdot P[X_1=x_1|Y=y] \cdot P[X_2=x_2|Y=y] \cdot P[X_3=x_3|Y=y]$ (4)

$P[Y=0|X_1=0, X_2=1, X_3=1] = P[Y=0] \cdot P[X_1=0|Y=0] \cdot P[X_2=1|Y=0] \cdot P[X_3=1|Y=0]$ (3)

$= \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{1}{27}$ (3)

(b) Laplace smoothing with k=1 gives

X_1	X_2	X_3	Y	
1	0	1	0	
0	0	1	0	given
1	1	0	0	
0	1	0	0	
0	1	1	0	added
0	0	0	1	
1	1	1	1	given
1	0	0	1	
0	1	0	1	
0	1	1	1	added

$P[Y=1|X_1=0, X_2=1, X_3=0]$ (5)

$= P[Y=1] \cdot P[X_1=0|Y=1] \cdot P[X_2=1|Y=1] \cdot P[X_3=0|Y=1]$

$= \frac{1}{2} \cdot \frac{3}{5} \cdot \frac{3}{5} \cdot \frac{3}{5}$ (5)

$= \frac{27}{250}$

Q2. [25 pts] Tikhonov Regularization

Let's take a look at a more complicated version of ridge regression called *Tikhonov regularization*. We use a regularization parameter similar to λ, but instead of a scalar, we use a real, square matrix $\Gamma \in \mathbb{R}^{d \times d}$ (called the *Tikhonov matrix*). Given a design matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and a vector of labels $\boldsymbol{y} \in \mathbb{R}^n$, our regression algorithm finds the weight vector $\boldsymbol{w}^* \in \mathbb{R}^d$ that minimizes the cost function

$$J(\boldsymbol{w}) = \|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|_2^2 + \|\Gamma\boldsymbol{w}\|_2^2$$

(a) [5 pts] Derive the normal equations for this minimization problem—that is, a linear system of equations whose solution(s) is the optimal weight vector $\boldsymbol{w}^*$. Show your work. (If you prefer, you can write an explicit closed formula for $\boldsymbol{w}^*$.)

(b) [5 pts] Give a simple, sufficient, and necessary condition on Γ (involving *only* Γ; not **X** nor $\boldsymbol{y}$) that guarantees that $J(\boldsymbol{w})$ has only one unique minimum $\boldsymbol{w}^*$. (To be precise, the uniqueness guarantee must hold for *all* values of **X** and $\boldsymbol{y}$ although the unique $\boldsymbol{w}^*$will be different for different values of **X** and $\boldsymbol{y}$.) (A sufficient but not necessary condition will receive part marks.)

(c) [10 pts] Recall the Bayesian justification of ridge regression. We impose an isotropic normal prior distribution on the weight vector—that is, we assume that $\boldsymbol{w} \sim \mathcal{N}(0, \sigma^2\mathrm{I})$. (This encodes our suspicion that small weights are more likely to be correct than large ones.) Bayes' Theorem gives us a posterior distribution $f(\boldsymbol{w}|\mathbf{X}, \boldsymbol{y})$. We apply maximum likelihood estimation (MLE) to estimate $\boldsymbol{w}$ in that posterior distribution, and it tells us to find $\boldsymbol{w}$ by minimizing $\|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|_2^2 + \lambda\|\boldsymbol{w}\|_2^2$ for some constant λ.
Suppose we change the prior distribution to an anisotropic normal distribution: $\boldsymbol{w} \sim \mathcal{N}(0, \Sigma)$. for some symmetric, positive definite covariance matrix Σ. Then MLE on the new posterior tells us to do Tikhonov regularization! What value of Γ does MLE tells us to use when we minimize $J(\boldsymbol{w})$? Give a one-sentence explanation of your answer.

(d) [5 pts] Suppose you solve a Tikhonov regularization problem in a two-dimensional feature space ($d = 2$) and obtain a weight vector $\boldsymbol{w}^*$ that minimizes $J(\boldsymbol{w})$. The solution $\boldsymbol{w}^*$ lies on an isocontour of $\|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|_2^2$and on an isocontour of $\|\Gamma\boldsymbol{w}\|_2^2$. Draw a diagram that plausibly depicts both of these two isocontours, in a case where Γ is not diagonal and $y \neq 0$. (You don't need to choose specific values of **X**, $\boldsymbol{y}$, or Γ; your diagram just needs to look plausible.) Your diagram must contain the following elements:

- The two axes (coordinate system) of the space you are optimizing in, with both axes labeled.
- The specified isocontour of $\|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|_2^2$, labeled.
- The specified isocontour of $\|\Gamma\boldsymbol{w}\|_2^2$, labeled.
- The point $\boldsymbol{w}^*$.

Q2.

(a)

The gradient of the cost function is

$$\nabla J(w) = 2X^\top(Xw - y) + 2\Gamma^\top\Gamma w.$$

Setting $\nabla J = 0$ gives the normal equations,

$$(X^\top X + \Gamma^\top\Gamma)w^* = X^\top y.$$

If the matrix is not singular, we can write the closed formula

$$w^* = (X^\top X + \Gamma^\top\Gamma)^{-1}X^\top y.$$

(b)

The following conditions are all equivalent to each other, and are all correct answers. $\Gamma^\top\Gamma$ is positive definite. Γ is invertible. Γ is nonsingular. Γ has full rank. Γ has rank d.

(c)

$\Gamma = \Sigma^{-1/2}$. (Answers that incorporate an extra scalar factor, like $\Gamma = \lambda\Sigma^{-1/2}$, are also acceptable—maybe even preferable.) Because the log posterior will include a term equal to some constant times $w^\top\Sigma^{-1}w$, which is equal to $\|\Gamma w\|^2$ if $\Gamma = \Sigma^{-1/2}$.

(d)

The diagram should contain: the two coordinate axes labeled w_1 and w_2; an ellipse or circle, **not** centered at the origin, labeled $\|Xw - y\|^2$; an ellipse, **not** axis-aligned and **not** circular, centered at the origin, labeled $\|\Gamma w\|^2$; a point w^* on both ellipses; and the two ellipses must touch each other tangentially without crossing each other.

Q3. [25 pts] Multiclass Bayes Decision Theory

Let's apply Bayes decision theory to 3-class classification. Consider a weather station that constantly receives data from its radar systems and must predict what the weather will be on the next day. Concretely:

- The input X is a scalar value representing the level of cloud cover, with only four discrete levels: 25, 50, 75, and 100 (the percentage of cloud cover).
- The station must predict one of three classes Y corresponding to the weather tomorrow. $Y = y_0$ means sunny, y_1 means cloudy, and y_2 means rainy.
- The priors for each class are as follows: $P(Y = y_0) = 0.5$, $P(Y = y_1) = 0.3$, and $P(Y = y_2) = 0.2$. The station has measured the cloud cover on the days prior to 100 sunny days, 100 cloudy days, and 100 rainy days. From these numbers they estimated the class-conditional probability mass functions $P(X|Y)$:

Prior-Day Cloud Cover(X)	Sunny, $P(X\|Y = y_0)$	Cloudy, $P(X\|Y = y_1)$	Rainy, $P(X\|Y = y_2)$
25	0.7	0.3	0.1
50	0.2	0.3	0.1
75	0.1	0.3	0.3
100	0	0.1	0.5

- We use an asymmetric loss. Let z be the predicted class and y the true class (label).

$$L(z, y) = \begin{cases} 0, & z = y \\ 1, & y = y_0, z \neq y_0 \\ 2, & y = y_1, z \neq y_1 \\ 4, & y = y_2, z \neq y_2 \end{cases}.$$

(a) [10 pts] Consider the constant decision rule $r_0(x) = y_0$, which *always* predicts y_0 (sunny). What is the average loss or equivalently risk $R(r_0)$ of the decision rule r_0? Your answer should be a number but show all your work.

(b) [15 pts] Derive the Bayes optimal decision rule $r^*(x)$—the rule that minimizes the risk $R(r^*)$.
Hint: Write down a table calculating $L(z, y_i)\, P(X|Y = y_i)\, P(Y = y_i)$, for each class y_i and each possible value of X (12 entries total), in the cases where the prediction z is wrong. Then figure out how to use it to minimize the risk R. This problem can be solved without wasting time computing $P(X)$.

(a)

$$\begin{aligned} R(r_0(x)) &= \sum_i \sum_x P(Y = y_i) L(r_0(x), y_i) P(X = x|Y = y_i) \\ &= P(Y = y_1) L(y_0, y_1) \sum_x P(X = x|Y = y_1) + P(Y = y_2) L(y_0, y_2) \sum_x P(X = x|Y = y_2) \\ &= 0.3 \cdot 2 \cdot 1 + 0.2 \cdot 4 \cdot 1 \\ &= 1.4. \end{aligned}$$

As r_0 does not actually vary with x at all, it is also acceptable to notice that we don't have to break down the risk by x:

$$\begin{aligned} R(r_0(x)) &= R(y_0) \\ &= \sum_i P(Y = y_i) L(y_0, y_i) \\ &= 0.3 \cdot 2 + 0.2 \cdot 4 \\ &= 1.4. \end{aligned}$$

(b)

The usual principle is to "pick the class with the biggest posterior probability," but in this case we also need to weight each class with an asymmetrical loss. We don't need to fully compute each posterior probability, because we don't need to know the value of the denominator $P(X)$. The following table shows $L(z, y_i)\, P(X|Y = y_i)\, P(Y = y_i)$, for each class y_i and each possible value of X, in the cases where the prediction z is wrong. (When z is correct, the loss is always zero.)

Prior-Day Cloud Cover (X)	Sunny (y_0)	Cloudy (y_1)	Rain (y_2)
25	**0.35**	0.18	0.08
50	0.10	**0.18**	0.08
75	0.05	0.18	**0.24**
100	0.00	0.06	**0.40**

The Bayes optimal decision rule always picks the highest loss-weighted posterior to minimize risk, so we pick the boldfaced classes (table 2) for each of the corresponding inputs X. Formally,

$$r(x) = \begin{cases} y_0 & x = 25, \\ y_1 & x = 50, \\ y_2 & x = 75 \text{ or } x = 100. \end{cases}$$

1. In this problem, you will see how to learn a binary Naive Bayes classifier. Consider the dataset below. You are looking to buy a car; the label is 1 if you are interested in the car and 0 if you aren't. There are three features: whether the car is red (your favorite color), whether the car is affordable, and whether the car is fuel-efficient.

Interested?	Red?	Affordable?	Fuel-Efficient?
1	1	1	1
0	0	1	0
0	0	1	1
1	0	0	0
0	0	1	1
0	0	1	1
1	1	1	1
1	1	0	1
0	0	0	0

(a) [5 pts] How many parameters do we need to learn?
(b) [10 pts] Estimate the parameters via MLE..
(c) [5 pts] If I see a car that is red, not affordable, and fuel-efficient, would the classifier predict that I would be interested in it?
(d) [5 pts.] Is there a problem with this classifier based on your calculations for the previous question? If so, how can we fix it?

a) 6 for $P(X|Y)$, 1 for $P(Y)$

b)

	$Y=1$	$Y=0$
Red?	$\frac{3}{4}$	0
Affordable?	$\frac{1}{2}$	$\frac{4}{5}$
Fuel-Efficient?	$\frac{3}{4}$	$\frac{3}{5}$

c) $P(Y=1|\text{red, not affordable, efficient}) \propto \frac{4}{9}\cdot\frac{3}{4}\cdot\frac{2}{4}\cdot\frac{3}{4}=\frac{1}{8}$
$P(Y=0|\text{red, not affordable, efficient}) \propto \frac{5}{9}\cdot 0\cdot\frac{1}{5}\cdot\frac{3}{5}=0$

d) If the car is red, the classifier will always predict I'm interested because $P(\text{not red}|Y=0)=0$. We can use a prior which prevents parameter estimates from being 0, i.e. adding 1 fake count for each feature/label combination.

5. (a) [5 pts] What are the four main tasks in machine learning?
(b) [5 pts] What is the corresponding machine learning task if want to estimate the price of a house based on a dataset. Indicate the underlined machine learning task if want to decide if an email is spam or not based on a dataset.

(a) Regression, classification, clustering, coding/dimensionality reduction
(b) Estimating the price of a house based on a dataset is a regression problem. Deciding if an email is spam or not based on a dataset is a classification problem.

3. In this problem, you will see how to learn a decision tree. Consider the following data set with three features X_1, X_2, X_3, and label Y.

(a) [20 pts] Suppose calculate the mutual information for each feature (X_1, X_2, X_3) to determine the root node for a Decision Tree trained on the data.. What feature should be split on at the root node?
(b) [10 points] Calculate what the next split should be.
(c) [5 pts.] Draw the resulting tree.

Outlook (X_1)	Temperature (X_2)	Humidity (X_3)	Play Tennis? (Y)
sunny	hot	high	no
overcast	hot	high	yes
rain	mild	high	yes
rain	cool	normal	yes
sunny	mild	high	no
sunny	mild	normal	yes
rain	mild	normal	yes
overcast	hot	normal	yes

(a)

$H(Y) = -\frac{6}{8} * \log_2(\frac{6}{8}) - \frac{2}{8} * \log_2(\frac{2}{8}) \approx 0.811$

For attribute X_1,

- $H(Y \mid X_1 = sunny) = -[\frac{1}{3} * \log_2(\frac{1}{3}) + \frac{2}{3} * \log_2(\frac{2}{3})] \approx 0.918$
- $H(Y \mid X_1 = rain) = 0$
- $H(Y \mid X_1 = overcast) = 0$

$\implies H(Y \mid X_1) = [\frac{3}{8} * 0.918 + \frac{3}{8} * 0 + \frac{2}{8} * 0] \approx 0.344$
$\implies I(Y; X_1) \approx 0.811 - 0.344 = 0.467$

For attribute X_2,

- $H(Y \mid X_2 = hot) = -[\frac{1}{3} * \log_2(\frac{1}{3}) + \frac{2}{3} * \log_2(\frac{2}{3})] \approx 0.918$
- $H(Y \mid X_2 = cool) = 0$
- $H(Y \mid X_2 = mild) = -[\frac{3}{4} * \log_2(\frac{3}{4}) + \frac{1}{4} * \log_2(\frac{1}{4})] \approx 0.811$

$\implies H(Y \mid X_2) = [\frac{3}{8} * 0.918 + \frac{1}{8} * 0 + \frac{4}{8} * 0.811] \approx 0.75$
$\implies I(Y; X_2) \approx 0.811 - 0.75 = 0.061$

For attribute X_3,

- $H(Y \mid X_3 = high) = -[\frac{1}{2} * \log_2(\frac{1}{2}) + \frac{1}{2} * \log_2(\frac{1}{2})] = 1$
- $H(Y \mid X_2 = normal) = 0$

$\implies H(Y \mid X_3) = [\frac{4}{8} * 1.0 + \frac{4}{8} * 0] = 0.5$
$\implies I(Y; X_3) \approx 0.811 - 0.5 = 0.311$

- Split on X_1 at the root node Since splitting on attribute X_1 gives the highest mutual information, the root node is X_1.

(b)

From the above part, as we can see that the sub-datasets $\mathcal{D}_{(X_1=rain)}$ and $\mathcal{D}_{(X_1=overcast)}$ are pure, there will be no further splitting on those and we will place a leaf node with label assignment decided by majority vote classifier. So, we need to split only on the sub-dataset $\mathcal{D}_{(X_1=sunny)}$. Now, we will use only $\mathcal{D}_{(X_1=sunny)}$ to estimate the probabilities for the next split.

$H(Y) = -\frac{1}{3} * \log_2(\frac{1}{3}) - \frac{2}{3} * \log_2(\frac{2}{3}) \approx 0.918$

For attribute X_2,

- $H(Y \mid X_2 = hot) = 0$
- $H(Y \mid X_2 = cool) = 0$
- $H(Y \mid X_2 = mild) = -[\frac{1}{2} * \log_2(\frac{1}{2}) + \frac{1}{2} * \log_2(\frac{1}{2})] = 1$

$\implies H(Y \mid X_2) = [\frac{2}{3} * 1.0 + \frac{1}{3} * 0] \approx 0.67$
$\implies I(Y; X_2) \approx 0.918 - 0.67 \approx 0.25$

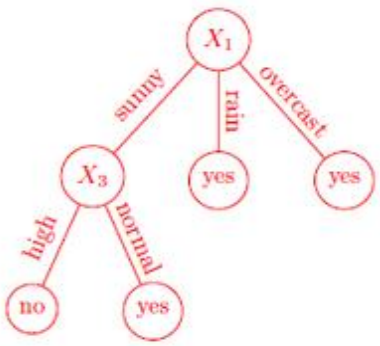
For attribute X_3,

- $H(Y \mid X_3 = high) = 0$
- $H(Y \mid X_3 = normal) = 0$

$\implies H(Y \mid X_3) = [\frac{2}{3} * 0 + \frac{1}{3} * 0] = 0$
$\implies I(Y; X_3) \approx 0.918$

We split using attribute X_3 as it gives the highest mutual information.

(c)



2. [25 pts]Suppose you are an avid BTS (a South Korean boy band that formed in 2010 and debuted in 2013) stan who monitors the social media accounts of each of the members. Suppose you wish to find the probability that a BTS member will post at any time of day. Over the three days period, you look on Instagram and find the following number of new posts: $x = [3, 4, 1]$. A fellow stan tells you that this comes from a Poisson distribution

$$p(x|\theta) = \frac{e^{-\theta}\theta^x}{x!}$$

Also, you are told that θ ~ Gamma(2, 2) — that is, its pdf is:

$$p(\theta) = \frac{1}{4}\theta e^{-\frac{\theta}{2}},\ \theta > 0$$

Calculate $\hat{\theta}_{MAP}$,

Note:

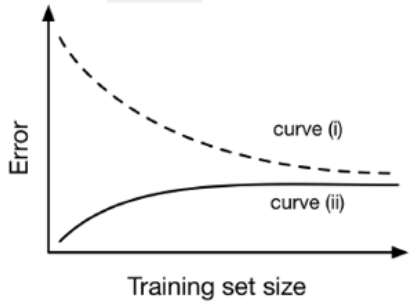
$$p(\mathcal{D}|\theta) = \frac{e^{-\theta}\theta^3}{3!}\frac{e^{-\theta}\theta^4}{4!}\frac{e^{-\theta}\theta^1}{1!}$$

$$\begin{aligned}
\hat{\theta}_{MAP} &= \arg\min_\theta -\log\left(p(\mathcal{D}|\theta)p(\theta)\right) \\
&= \arg\min_\theta -\log\left(\frac{e^{-\theta}\theta^3}{3!}\frac{e^{-\theta}\theta^4}{4!}\frac{e^{-\theta}\theta^1}{1!}\frac{1}{\Gamma(2)2^2}\theta^{(2-1)}e^{\frac{-\theta}{2}}\right) \\
&= \arg\min_\theta -\log e^{-3\theta}\theta^8\theta^{(2-1)}e^{\frac{-\theta}{2}} \\
&= \arg\min_\theta -\log e^{-3\theta-\frac{\theta}{\beta}}\theta^{8+\alpha-1} \\
&= \arg\min_\theta -((-3\theta-\frac{\theta}{2}) + (8+2-1)\log(\theta))
\end{aligned}$$

Setting the derivative equal to zero yields

$$\begin{aligned}
0 &= -3 - \frac{1}{2} + \frac{(7+2)}{\theta} \\
\implies \theta_{MAP} &= \frac{7+2}{3+\frac{1}{2}} = 2.57142857143
\end{aligned}$$

4. In this problem, we will consider the effect of training dataset size n on a logistic regression classifier with d features. The classifier is trained by optimizing the conditional log-likelihood. The optimization procedure stops if the estimated parameters perfectly classify the training data, or they converge. The following plot shows the general trends in training and testing error as we increase the sample size $n = |S|$



(a) [10 pts] Which curve represents the training error? Provide 1-2 sentences of justification.
(b) [5 pts] In one word, what does the gap between the two curves represent?

a) Curve (ii) is the training set. Training error increases as the training set increases in size (more points to account for). However, the increase tapers out when the model generalizes well. Evidently, curve (i) is testing, since larger training sets better form generalized models, which reduces testing error.

b) overfitting

2. In this problem, you will see how to learn a binary logistic regression classifier. Consider the dataset below, where Y is the class label and Xs are attributes

Y	X_1	X_2	X_3
1	1	2	1
1	1	1	-1
0	1	-2	1

(e) [5 pts] What is classification error $J(\boldsymbol{\theta})$ if the initial weight vector for the attributes is equal to $\boldsymbol{\theta} = [-2,2,1]^T$

(f) [5 pts] Calculate $\frac{\partial J^{(1)}(\theta_1)}{\theta_1}$, $\frac{\partial J^{(1)}(\theta_2)}{\theta_2}$, $\frac{\partial J^{(1)}(\theta_3)}{\theta_3}$ for the first training sample (first row). Note that $\sigma(3) \approx 0.95$

(g) [5 pts] Calculate $\frac{\partial J^{(1)}(\theta_1)}{\theta_1}$, $\frac{\partial J^{(1)}(\theta_2)}{\theta_2}$, $\frac{\partial J^{(1)}(\theta_3)}{\theta_3}$ for the second training sample (first row). Note that $\sigma(-1) \approx 0.25$

(h) [5 pts.] Assuming we are doing stochastic gradient descent with a learning rate of 1.0, what are the updated parameters $\boldsymbol{\theta}$ if we update $\boldsymbol{\theta}$ using the second training example?

(i) [5 pts] What is the new $J(\theta)$ after doing the above update? Should it decrease or increase?

(j) [5 pts] Given a test example where $(X_1 = 1, X_2 = 3, X_3 = 4)$, what will the classifier output following this update?

(a)

$$J(\boldsymbol{\theta}) = -\tfrac{1}{3}[\log(\sigma(3)) + \log(\sigma(-1)) + \log(1 - \sigma(-5))] \approx 0.46$$

(b)

$$\frac{\partial J^{(1)}(\boldsymbol{\theta})}{\partial \theta_1} = (\sigma(3) - 1)1 = -0.05$$
$$\frac{\partial J^{(1)}(\boldsymbol{\theta})}{\partial \theta_2} = (\sigma(3) - 1)2 = -0.10$$
$$\frac{\partial J^{(1)}(\boldsymbol{\theta})}{\partial \theta_3} = (\sigma(3) - 1)1 = -0.05$$

5

(c)

$$\frac{\partial J^{(2)}(\boldsymbol{\theta})}{\partial \theta_1} = (\sigma(-1) - 1)1 = -0.75$$
$$\frac{\partial J^{(2)}(\boldsymbol{\theta})}{\partial \theta_2} = (\sigma(-1) - 1)1 = -0.75$$
$$\frac{\partial J^{(2)}(\boldsymbol{\theta})}{\partial \theta_3} = (\sigma(-1) - 1) - 1 = 0.75$$

5

(d)

$$\begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix} - 1 \begin{bmatrix} -0.75 \\ -0.75 \\ 0.75 \end{bmatrix} = \begin{bmatrix} -1.25 \\ 2.75 \\ 0.25 \end{bmatrix}$$

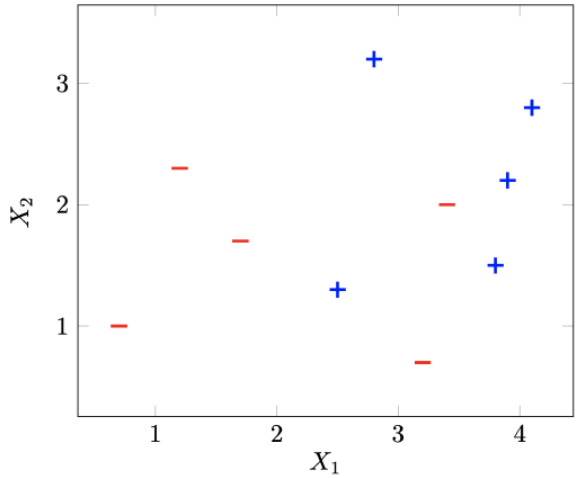
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(e) J=0.09, it should decrease for logistic classifier to learn. 5

(f)

$$\sigma(\theta^T X) > 0.5 \implies Y = 1$$

5



3. In this problem, you will see how to learn a decision tree. Consider the following data set with two features X_1, X_2, and label Y, where '−' marks represent $Y = 0$ and '+' marks represent $Y = 1$.

(a) [6 pts] What is the entropy of Y in bits?
(b) [6 pts] What is the mutual information if we are splitting on $X_2 < 2.5$?
(c) [6 pts] Using error rate as the splitting criteria, what is the best choice for the first binary split?
(d) [6 pts] If the tree only has one binary split (i.e., a decision stump), what would be the lowest training error rate we can get?
(e) [6 pts] Is it possible to have a decision tree with zero training error for this dataset? Assume only binary splits and attributes selected with replacement.

(a) -[(1/2)log(1/2) + (1/2)log(1/2)] = 1 bit
(b)

$$1 - 0.8 \cdot (-5/8 \cdot \log_2(5/8) - 3/8 \cdot \log_2(3/8)) = 0.23645$$

(c)

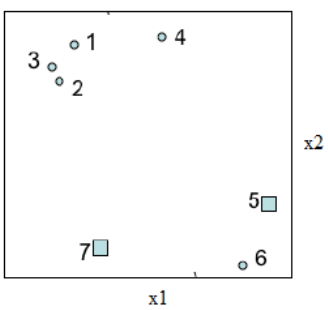
$$X_1 < 2 \text{ and } X_1 \geq 2$$

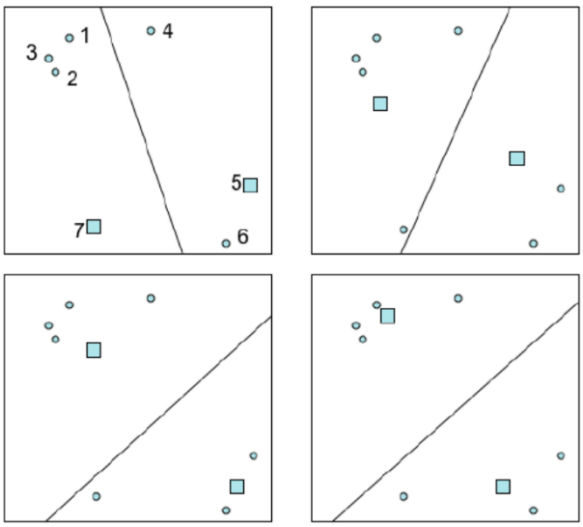
(d)

$$0.2 + -0.001$$

(e) Yes, it is possible.

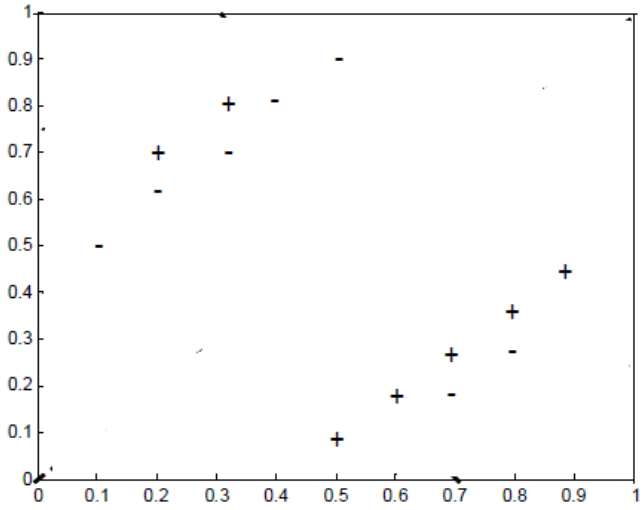
1. [25 pts] Perform K-means clustering algorithm (there are two classes) on the two-dimensional dataset given below. Circles are data points and there are two initial cluster centers, at data points 5 and 7. Draw the cluster centers (as squares) and the decision boundaries that define each cluster. If no points belong to a particular cluster, assume its center does not change. Use as many of the pictures as you need for convergence.



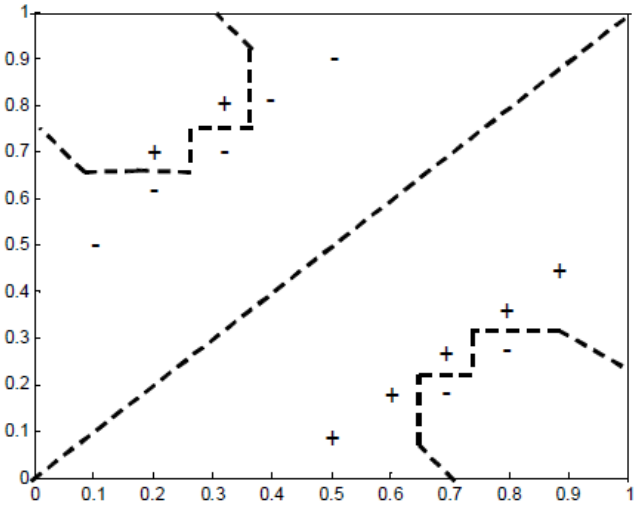


2. In the following questions you will consider a k-nearest neighbor classifier using Euclidean distance metric on a binary classification task. We assign the class of the test point to be the class of the majority of the k-nearest neighbors. Note that a point can be its own neighbor. The 2-D dataset is given below. Points represented by plus (+) sign belong to one class and those represented by minus (-) sign belong to another class.

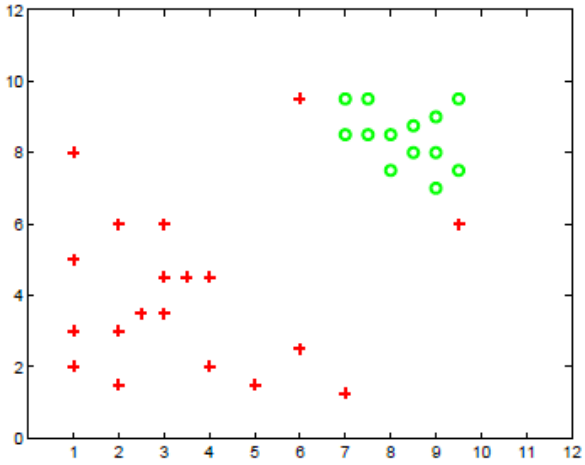
(a) [5 pts] What value of k minimizes the training set error for this dataset? What is the resulting training error?
(b) [10 pts] Why might using too large values k be bad in this dataset? Why might too small values of k also be bad?
(c) [5 pts] What value of k minimizes leave-one-out cross-validation error for this dataset? What is the resulting error?
(d) [5 pts] Sketch the 1-nearest neighbor decision boundary for this dataset.

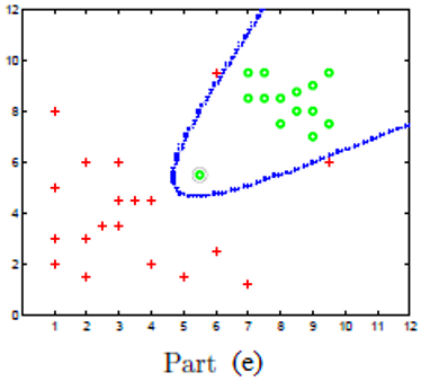


(a) Note that a point can be its own neighbor. So, k = 0 minimizes the training set error. The error is 0.
(b) Too big k (k = 13) misclassifies every datapoint (using leave one out cross validation). Too small k leads to overfitting
(c) k = 5 or k = 7 minimizes the leave-one-out cross-validation error. The error is 4/14.
(d)



The goal of this problem is to correctly classify test data points, given a training data set. You have been warned, however, that the training data comes from sensors which can be error-prone, so you should avoid trusting any specific point too much. For this problem, assume that we are training an SVM with a quadratic kernel– that is, our kernel function is a polynomial kernel of degree 2. You are given the data set presented in the figure below. The slack penalty C will determine the location of the separating hyperplane. Please answer the following questions qualitatively. Give a one sentence answer/justification for each and draw your solution.





Part (e)

(a) [5 pts] Where would the decision boundary be for very large values of C (i.e., $C \rightarrow \infty$)? (remember that we are using an SVM with a quadratic kernel.) Draw on the figure below. Justify your answer.

(b) [5 pts] For $C \approx 0$, indicate in the figure , where you would expect the decision boundary to be? Justify your answer.

(c) [5 pts] Which of the two cases above would you expect to work better in the classification task? Why?

(d) [5 pts] Draw a data point which will not change the decision boundary learned for very large values of C. Justify your answer.

(e) [5 pts] Draw a data point which will change the decision boundary learned for very large values of C. Justify your answer.

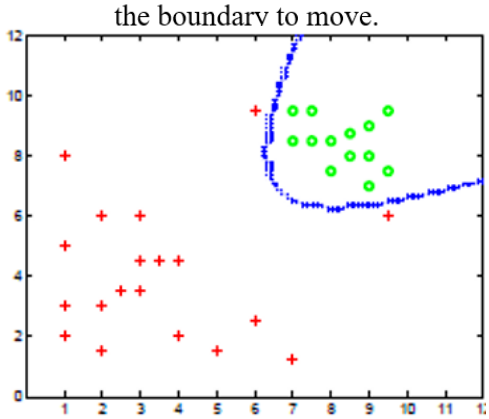
(a) For large values of C, the penalty for misclassifying points is very high, so the decision boundary will perfectly separate the data if possible. See below for the boundary learned using C = 100000.

(b) The classifier can maximize the margin between most of the points, while misclassifying a few points, because the penalty is so low. See below for the boundary learned using C = 0.00005.

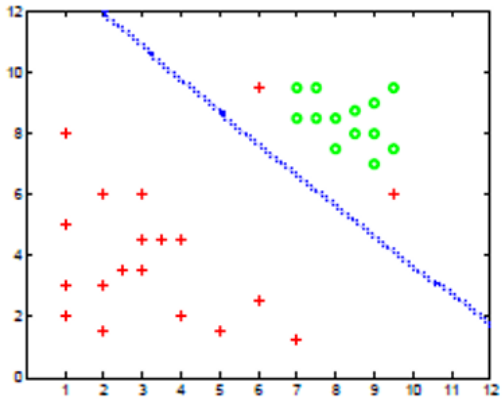
(c) We were warned not to trust any specific data point too much, so we prefer the solution where C ≈ 0, because it maximizes the margin between the dominant clouds of points.

(d) We add the point circled below, which is correctly classified by the original classifier, and will not be a support vector.

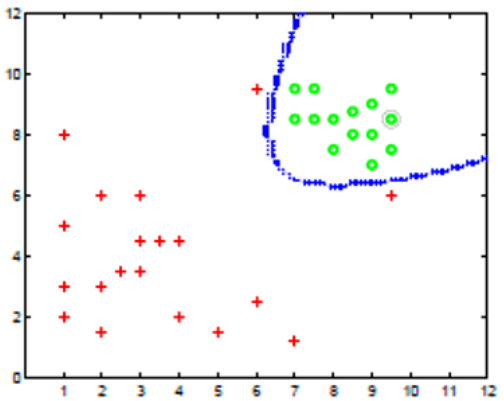
(e) Since C is very large, adding a point that would be incorrectly classified by the original boundary will force the boundary to move.



Part (a)



Part (b)



Part (d)

4. In this problem, the following short questions should be answered with at most two sentences, and/or a picture. For the (true/false) questions, answer true or false. If you answer true, provide a short justification, if false explain why or provide a small counterexample.

   (a) [5 pts] Discuss whether MAP estimates are less prone to overfitting than MLE.
   (b) [5 pts] Consider a classification problem with two classes and n binary attributes. How many parameters would you need to learn with a Naive Bayes classifier? How many parameters would you need to learn with a Bayes optimal classifier?
   (c) [5 pts] For an SVM, if we remove one of the support vectors from the training set, does the size of the maximum margin decrease, stay the same, or increase for that dataset?
   (d) [5 pts] **true/false** In n-fold cross-validation each data point belongs to exactly one test fold, so the test folds are independent. Are the error estimates of the separate folds also independent? So, given that the data in test folds i and j are independent, are e_i and e_j , the error estimates on test folds i and j, also independent?
   (e) [5 pts] **true/false** There is an a priori good choice of n for n-fold cross-validation.

(a) Usually, MAP is less prone to overfitting than MLE. MAP introduces a prior over the parameters. So, given prior knowledge, we can bias the values of the parameters. MLE on the other hand just returns the most likely parameters. Whether MAP is really less prone to overfitting depends on which prior is used – an uninformative (uniform) prior can lead to the same behavior as MLE.
(b) NB has $1 + 2n$ parameters — prior $P(y = T)$ and for every attribute x_i, we have $p(x_i = T|y_i = T)$ and $p(x_i = T|y_i = F)$. For optimal Bayes for every configuration of attributes we need to estimate $p(y|x)$. This means we have 2^n parameters.
(c) [5 pts] The margin will either increase or stay the same, because support vectors are the ones that hold the marging from expanding. Here is an example of increasing margin. Suppose we have one feature $x \in \mathbb{R}$ and binary class y. The dataset consists of 3 points:

$$(x_1, y_1) = (-1, -1), \quad (x_2, y_2) = (1, 1), \quad (x_3, y_3) = (3, 1).$$

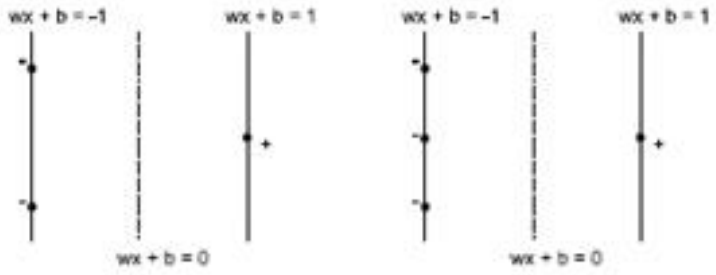


Figure 1: Example of SVM margin remaining the same after one support vector is deleted

For standard SVM with slacks the optimal separating hyperplane $wx+b=0$ has parameters

$$w = 1, b = 0$$

corresponding to the margin of $\frac{2}{|w|} = 2$. The support vectors are x_1 and x_2. If we remove (x_2, y_2) from the dataset, new optimal parameters for the separating hyperplane will be

$$w = \frac{1}{2}, b = \frac{-1}{2}$$

for the new margin of $\frac{2}{|w|} = 4$.
If there are redundant support vectors, the margin may stay the same - see Fig. 1

(d) False. Since a data point appears in multiple folds the training sets are dependent and thus test fold error estimates are dependent
(e) False. We do not know the relation between sample size and accuracy. High n increases correlation in training set and decreases variance of estimates. How much depends on the data and the learning method